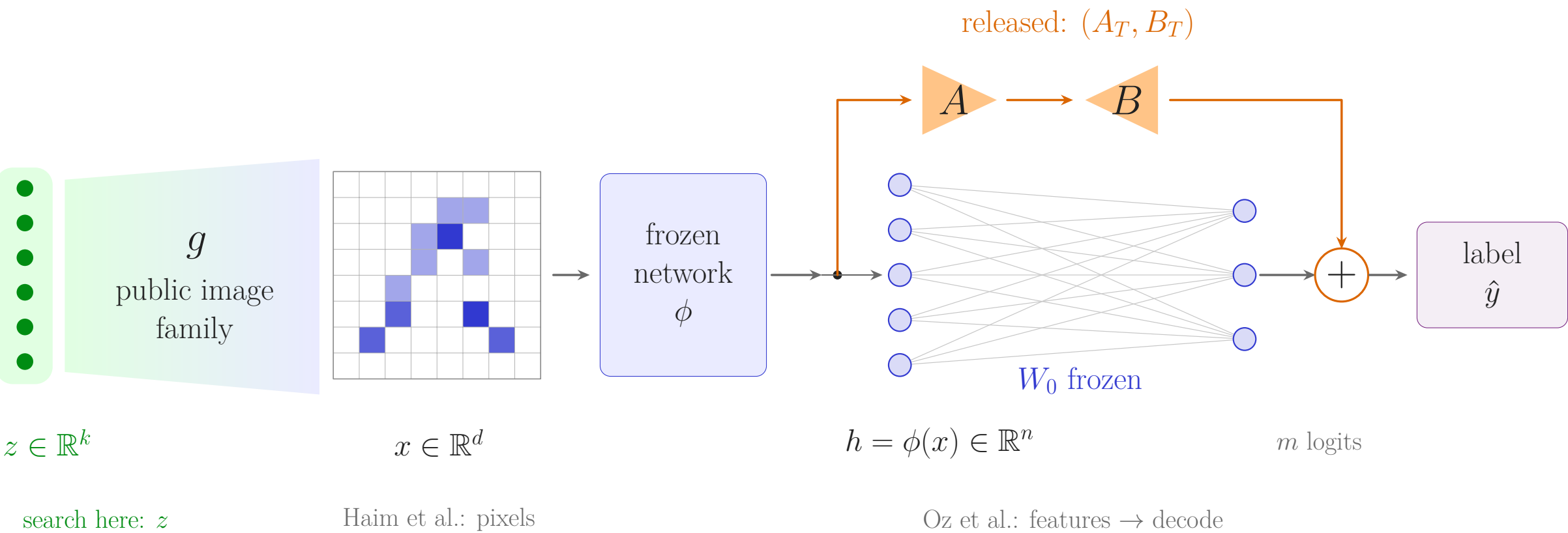
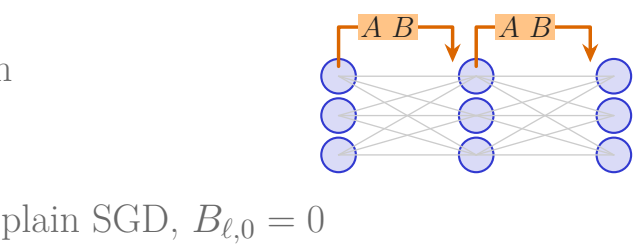


A public image family before the network



$$x = g(z), \quad f(x) = (W_0 + BA)\phi(x), \quad k \ll d$$

Deeper adapters
conditional extension

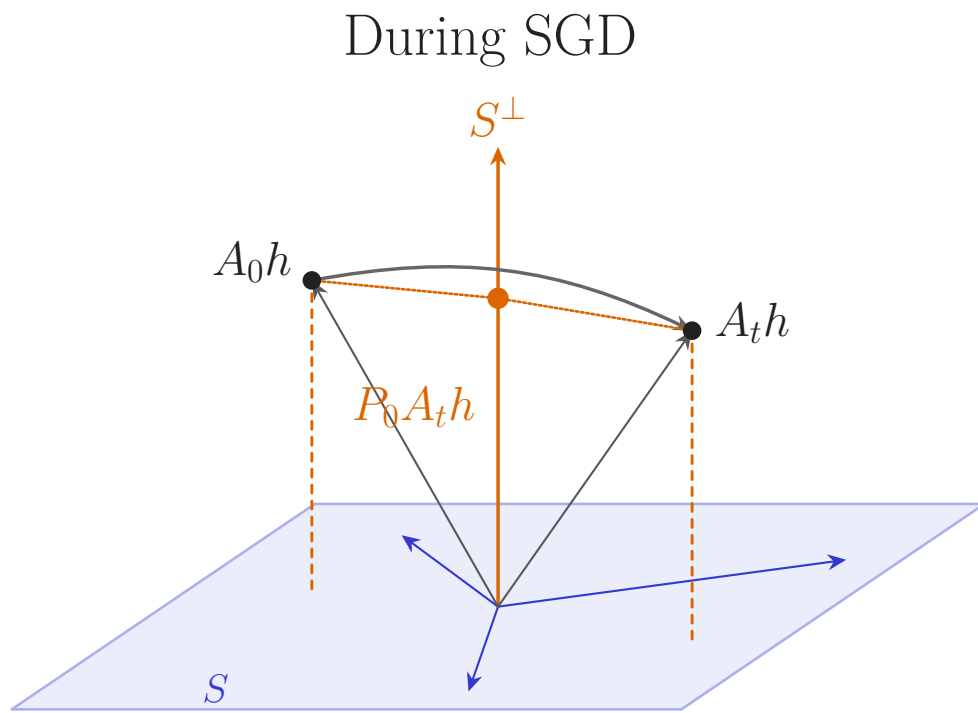


common trajectory/base span + full recording

$$\Phi_{\ell,t}(x_i), \Phi_{\ell}^0(x_i) \in \mathcal{U}_{\ell}, \quad \dim \mathcal{U}_{\ell} < r_{\ell}$$

$$\text{rank } B_{\ell,T} = \dim(A_{\ell,0}\mathcal{U}_{\ell})$$

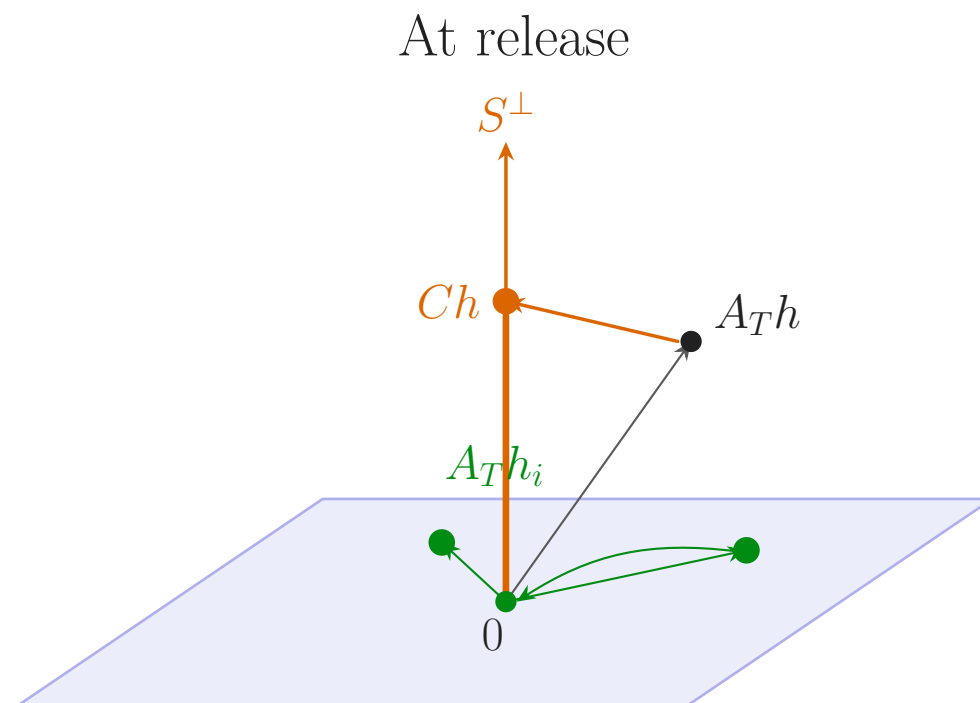
The preserved subspace and the certificate



$$S = \text{col}(A_0 H)$$

$$B_t P_0 = 0$$

$$P_0 A_t = P_0 A_0$$



$$S = \text{row}(B_T), \quad \dim S = q$$

$$C = P_0 A_T, \quad P_0 = P_{S^\perp}$$

$$C\phi(x_i) = 0, \quad \text{rank } C = r - q$$

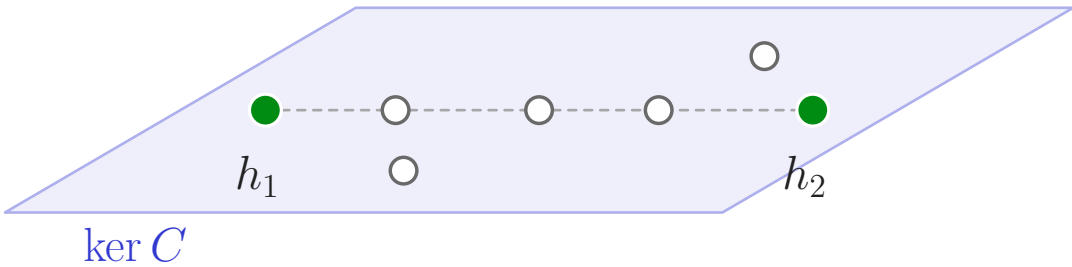
$$H = [\phi(x_1) \ \cdots \ \phi(x_N)] \text{ fixed,} \quad \text{rank } B_T = q = \text{rank } H < r \leq n$$

plain SGD; $B_0 = 0$; generic independent A_0 ; no weight decay

The family can separate compatible features

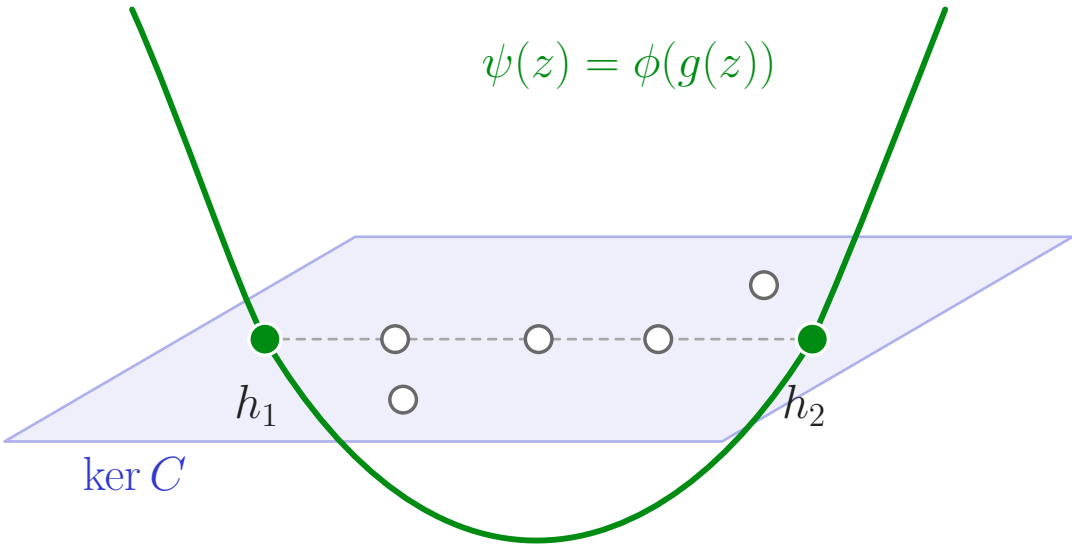
Feature-space schematic

Certificate alone



compatible features need not identify images

Certificate + family



separation is a property to establish

$$\underbrace{g(z) = \mu + Uz}_{\text{PCA: affine in image space}}$$

$$\underbrace{\psi(z) = \phi(\mu + Uz)}_{\text{can be curved in feature space}}$$

$$k < r - q, \qquad \{z : \psi(z) \in \text{col } H\} = \{z_1, \dots, z_N\}$$

fixed smooth family + span separation + exact-certificate conditions: generic identification

Motorcycles

CIFAR-100

$r = 64$ $k = 32$

Ground truth
original image



PCA input
used for training



Reconstruction



8/8 training inputs recovered

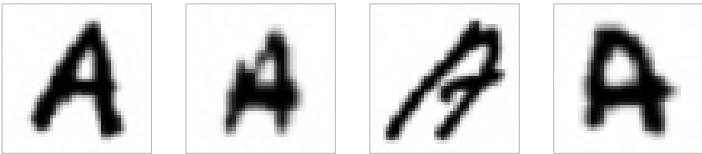
135/200 successful starts

Reconstructions

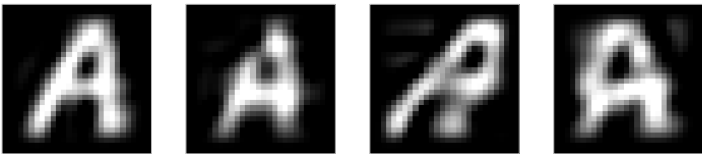
Letters

8/8 PCA targets; four shown

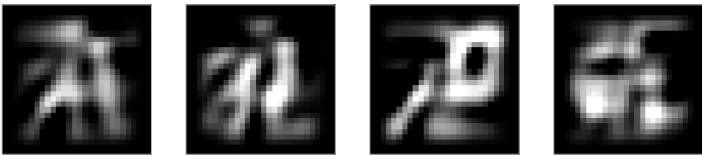
Original



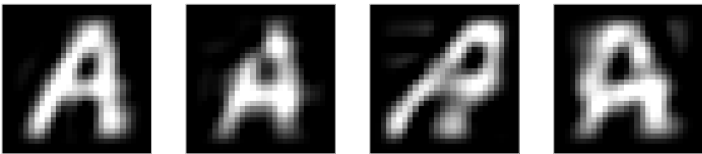
PCA training target



NTK output



Certificate output



Training: public PCA projections

CIFAR-100 motorcycles

8/8 PCA targets

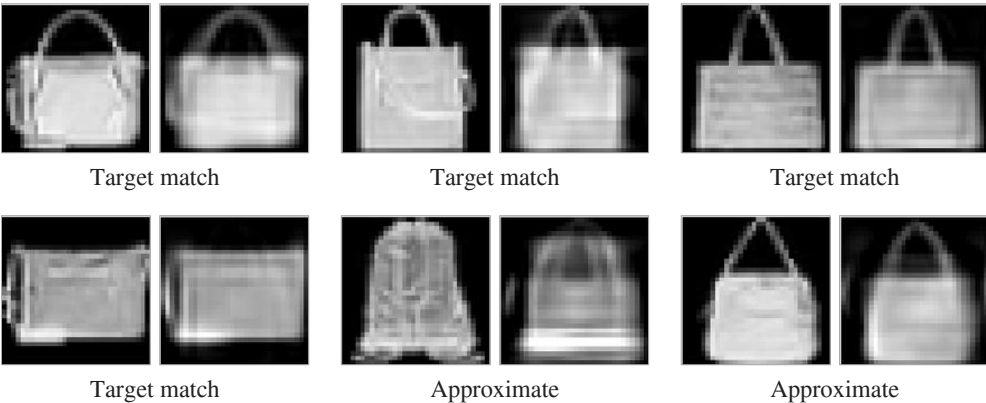
Original / reconstruction



Fashion-MNIST bags

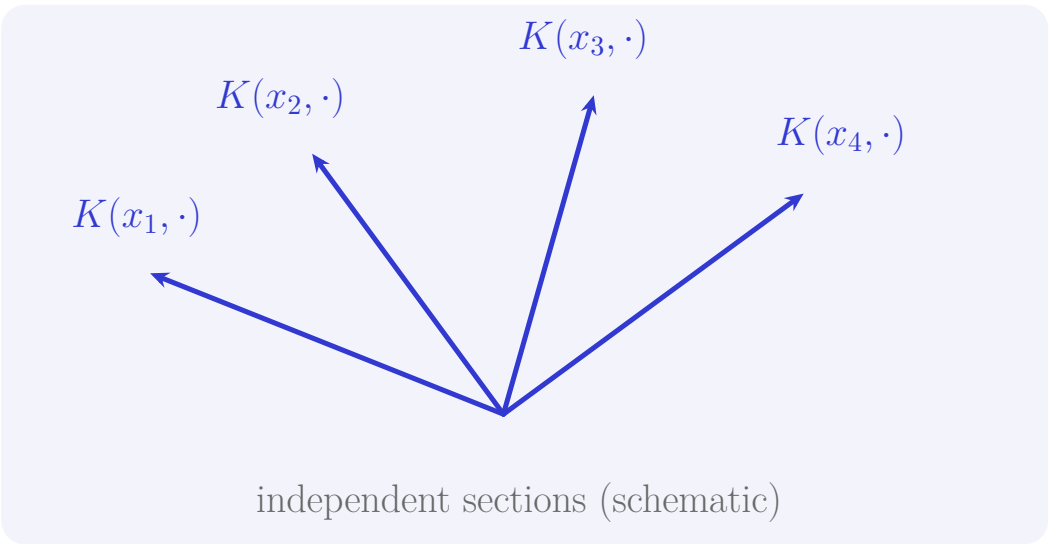
4/8 PCA targets + 2 approximations shown

Original / reconstruction



What changes in the NTK argument?

Universal-kernel guarantee

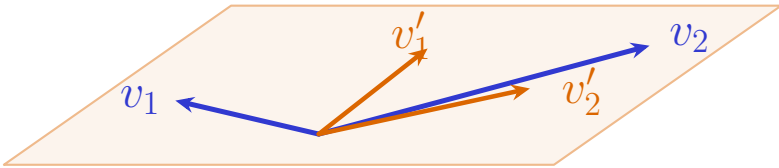


$$\sum_j a_j K(x_j, \cdot) = 0 \implies a_j = 0$$

distinct x_j

LoRA head at initialization

$$K^c(x, x') = \langle A_0 \phi(x), A_0 \phi(x') \rangle$$
$$\text{rank } K^c \leq r$$



$$V = [A_0 \phi(x_1) \cdots A_0 \phi(x_N)]$$

alternative bases; same free-coefficient fit

$$\Lambda V^\top = (\Lambda M^{-\top})(VM)^\top$$

$$T = 1 : \quad A_1 = A_0, \quad B_1 = -\eta D_0 (A_0 H)^\top$$

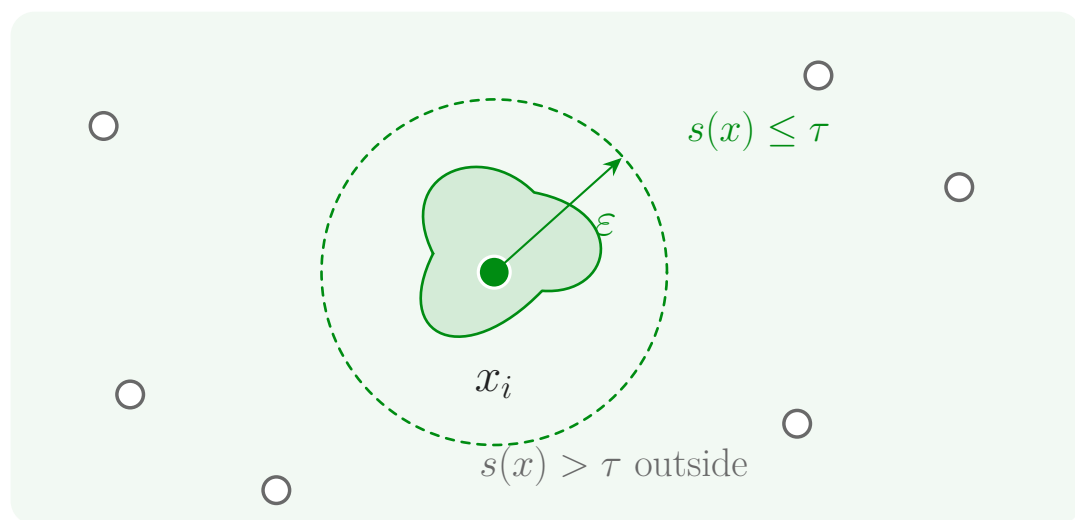
free coefficients; invertible M ; realizable candidate features required

Loo et al., Thm. 1: infinite width, unit sphere, nonzero training coefficients

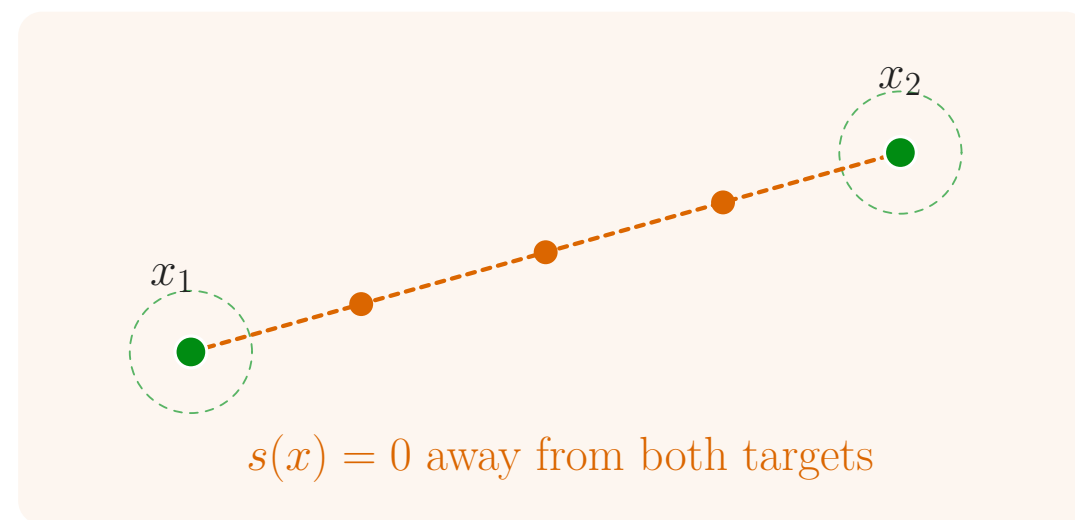
When can a low score be trusted?

$$s(x) = \frac{\|C\phi(x)\|}{\|A_T\phi(x)\|}, \quad \mathcal{X} = g(\mathcal{Z})$$

The desired separation



Affine-input counterexample



ϕ affine; admissible blends with nonzero denominator

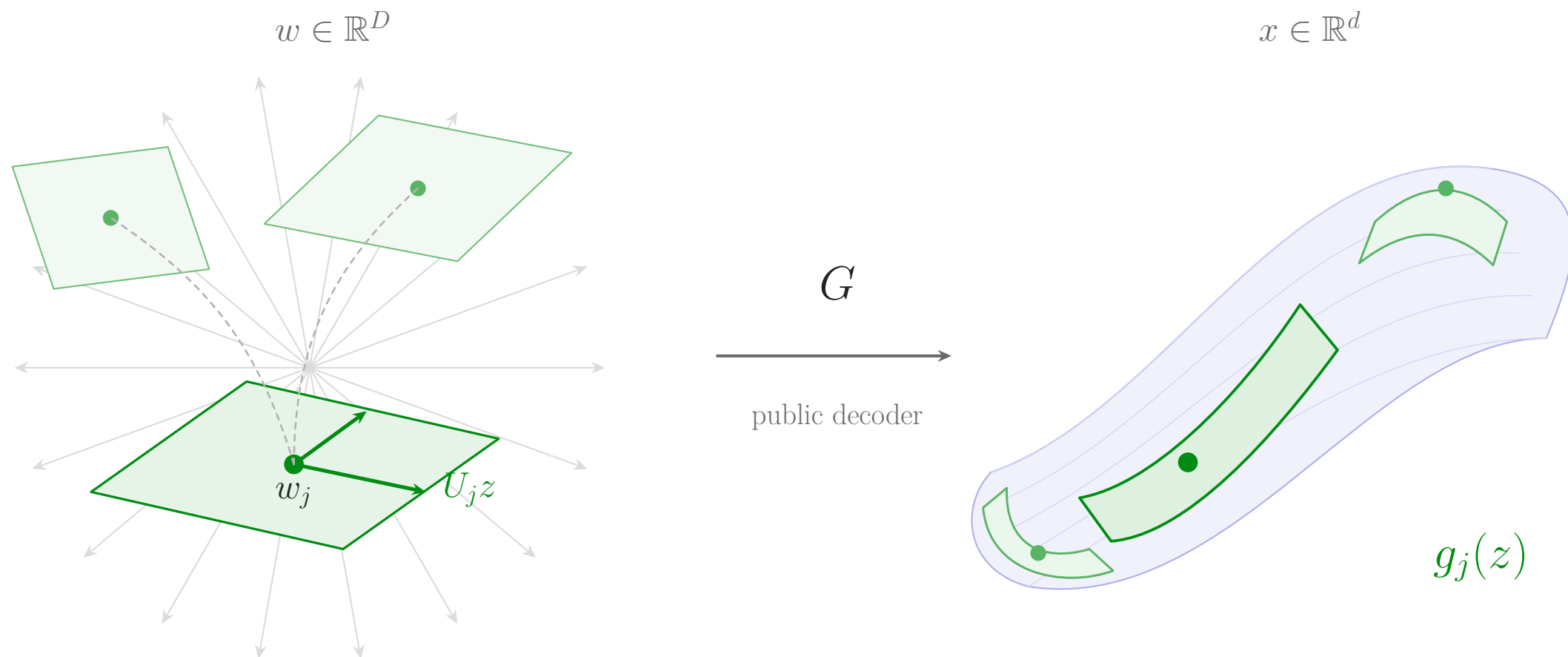
$$\gamma_{\mathcal{X}}(\epsilon) = \inf_{\substack{x \in \mathcal{X} \\ \min_i \|x - x_i\| > \epsilon}} s(x)$$

$$s(x) \leq \tau < \gamma_{\mathcal{X}}(\epsilon) \implies \min_i \|x - x_i\| \leq \epsilon$$

$x \in \mathcal{X}, \quad \|A_T\phi(x)\| > 0$ target: a positive wrong-image score floor on the allowed family

Local image families

Proposal



$$g_j(z) = G(w_j + U_j z)$$

$$z \in \mathbb{R}^k, \quad k < r - q, \quad k \ll D, \quad \|z\| \leq \rho_j$$

$$U_j \in \mathbb{R}^{D \times k}$$